Existing signal synchronization methods have limitations. PLLs and DLLs introduce different types of jitter and complex logic while modifying the phase of the original signals, and length matching introduces more parasitic effects and is limited to PCB design. My goal was to build a low-cost, simple-logic prototype to self-synchronize multiple low-amplitude input signals while preserving the original signal integrity.

Signals are amplified by non-inverting and inverting amplifiers. When both input signals' magnitude >20mV, my prototype will generate two logic-high signals, and the connected AND logic gate will output a high signal to turn on two MOSFETs, allowing the signals to pass simultaneously.

I used an oscilloscope to generate ±400mV signals and measure the signals' synchronization delay. I measured 120 trials at 1-3kHz for square waves and 10kHz for sine waves. The average synchronization delay was <=4ns delay with 96.67% reliability. The current bandwidth was limited to <=10kHz by the amplifier; breadboard parasitic effects affected the testing accuracy. The bandwidth can exceed 100MHz by using higher bandwidth components and a customized PCB. Through cascading, it can synchronize more than two signals. The method is new and low-cost. Compared to other methods, the prototype provides an auto-synchronization method for low-amplitude signals as small as 20mV without modifying the original signals' phase, frequency, or amplitude. It can reduce the backend controller computational load and improve real-time responsiveness. It can be applied in automotive, industrial, and medical fields and used for educational purposes.